CBSE Class 7 · Mathematics (Ganita Prakash, Part 2)

Geometric Twins — Answer Key

Congruence of figures and triangles · 30 questions

"Corresponding parts" means corresponding parts of congruent triangles (CPCT). Students may write equivalent reasons in their own words.

Section A — Congruent Figures and Corresponding Parts

1. (a) Yes — segments of equal length are congruent.
(b) Yes — squares with equal sides fit exactly on each other.
(c) Yes — circles with equal radii are congruent.
(d) No — e.g. a 3 cm × 4 cm rectangle and a 2 cm × 6 cm rectangle both have area 12 cm² but do not match.
(e) Yes — angles with equal measure are congruent.
(f) Yes — each side is 18 ÷ 3 = 6 cm, so all three sides match (SSS).

2. (a) P and Q are congruent (Q is P turned through a quarter turn). R and S are congruent (both are right triangles with legs 3 units and 2 units).
(b) No. T is a 4 × 1 rectangle and U is a 2 × 2 square; both have area 4 square units but they have different shapes, so one cannot be placed exactly on the other.

3. AB = PQ, BC = QR, CA = RP; ∠A = ∠P, ∠B = ∠Q, ∠C = ∠R.

4. (a) LM = XY = 6 cm (b) ∠L = ∠X = 50° (c) ∠M = ∠Y = 60°
(d) ∠N = ∠Z = 180° − 50° − 60° = 70°

5. (a) False — triangles with the same angles can have different sizes (AAA is not a criterion).
(b) True — congruent figures cover each other exactly.
(c) False — a 4 × 1 rectangle and a 2 × 2 square have equal area but are not congruent.
(d) False — one more side (or acute angle) must also be equal (RHS needs hypotenuse and one side).
(e) False — e.g. triangles with sides 3, 4, 5 cm and 4, 4, 4 cm both have perimeter 12 cm.
(f) True — a figure fits exactly on itself.

6. (a) Yes, by SSS: the side lengths 4 cm, 5 cm, 6 cm occur in both triangles.
(b) No. Match equal sides: AB = RP = 4, BC = PQ = 5, CA = QR = 6. So A ↔ R, B ↔ P, C ↔ Q. Correct statement: **ΔABC ≅ ΔRPQ**.

Section B — Criteria for Congruence (SSS, SAS, ASA, AAS, RHS)

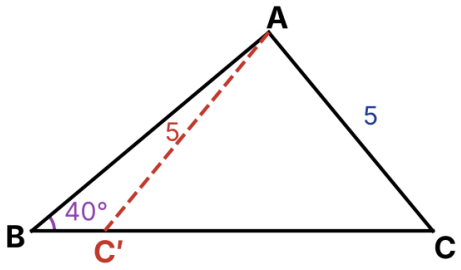
7. Congruent by **SAS**: AB = QR = 5 cm, ∠B = ∠Q = 50°, BC = QP = 7 cm. **ΔABC ≅ ΔRQP**.

8. Congruent by **ASA**: ∠L = ∠X = 60°, LM = XY = 6 cm, ∠M = ∠Y = 45°. **ΔLMN ≅ ΔXYZ**.

9. Congruent by **AAS**: ∠D = ∠S = 70°, ∠E = ∠T = 45°, and the side EF = TU = 5 cm (opposite the 70° angle). **ΔDEF ≅ ΔSTU**. (Also possible: third angles ∠F = ∠U = 65°, then ASA.)

10. Congruent by **RHS**: ∠B = ∠Q = 90°, hypotenuse AC = RP = 10 cm, side AB = RQ = 6 cm. **ΔABC ≅ ΔRQP**.

11. (a) No, the triangles are not congruent.
(b) BC is much longer than QR (BC ≈ 7.8 cm, QR ≈ 1.4 cm). The 40° angle is **not** between the two equal sides, so the side of 5 cm can swing to two different positions on the base line (C and C′ in the figure below). Two different triangles are possible, so SSA does not fix a unique triangle and is not a congruence criterion.



12. No. Only the angles are equal; DE is clearly longer than AB, so the triangles have the same shape but different sizes. Equal angles alone (AAA) do not guarantee congruence.

13. (a) AB = AD (given), CB = CD (given), AC = AC (common). So ΔABC ≅ ΔADC by **SSS**.
(b) ∠BAC = ∠DAC (corresponding parts), and together they make 80°, so each is **40°**.
(c) Yes. ∠ABC and ∠ADC are corresponding parts of the congruent triangles.

14. (a) ∠AOC = ∠BOD, because they are vertically opposite angles.
(b) OA = OB, ∠AOC = ∠BOD, OC = OD. So ΔAOC ≅ ΔBOD by **SAS**.
(c) BD = AC = **4.5 cm** (corresponding parts).
(d) ∠OBD (since A ↔ B, O ↔ O, C ↔ D).

Section C — Using Congruence to Find Unknown Parts

15. P ↔ S, Q ↔ T, R ↔ U. y = ST = PQ = **7 cm**; w = ∠S = ∠P = **55°**;
z = ∠U = ∠R = 180° − 55° − 65° = **60°**.

16. (a) **SAS**: QA = QB = 4 m, ∠PQA = ∠PQB = 90° (the pole is vertical), PQ = PQ (common).
(b) PB = PA = **5 m** (corresponding parts).
(c) Yes, they are corresponding angles of the congruent triangles.

17. (a) ∠DCA = ∠BAC — alternate angles, since AB ∥ DC and AC is a transversal.
(b) AB = CD (given), ∠BAC = ∠DCA, AC = CA (common). So ΔABC ≅ ΔCDA by **SAS**.
(c) AD = CB = **6 cm**.
(d) ∠ABC = ∠CDA (corresponding parts).

18. (a) ∠XPM = ∠YQM (alternate angles, PX ∥ QY); ∠PMX = ∠QMY (vertically opposite angles). [Also ∠PXM = ∠QYM, alternate angles.]
(b) ∠XPM = ∠YQM, PM = QM (M is the midpoint), ∠PMX = ∠QMY. So ΔPMX ≅ ΔQMY by **ASA**.
(c) MX = MY, so MX = 10 ÷ 2 = **5 cm**.
(d) QY = PX = **3.2 cm**.

19. Angles opposite equal sides are equal, so ∠B = ∠C. ∠B + ∠C = 180° − 40° = 140°. So ∠B = ∠C = **70°**.

20. Sides opposite equal angles are equal, so AC = AB = **7 cm**. ∠A = 180° − 65° − 65° = **50°**.

21. ∠PRQ = 180° − 115° = **65°** (linear pair). ∠PQR = ∠PRQ = **65°** (PQ = PR). ∠QPR = 180° − 65° − 65° = **50°**.

22. (a) Each angle is **60°**: all sides are equal, so all angles are equal, and 180° ÷ 3 = 60°.
(b) ∠ADB = ∠ADC = 90°, hypotenuse AB = AC, AD = AD (common). So ΔABD ≅ ΔACD by **RHS**.
(c) BD = DC (corresponding parts), so BD = DC = 8 ÷ 2 = **4 cm**.
(d) In ΔABD: ∠BAD = 180° − 90° − 60° = **30°**.

23. (a) AB = AC (given), BD = CD (D is the midpoint), AD = AD (common). ΔABD ≅ ΔACD by **SSS**.
(b) ∠BAD = ∠CAD, so ∠BAD = 70° ÷ 2 = **35°**.
(c) ∠ADB = ∠ADC (corresponding parts) and they form a linear pair (sum 180°). So ∠ADB = **90°**.
(d) ∠ABD = 180° − 35° − 90° = **55°**. (Check: (180° − 70°) ÷ 2 = 55°.)

Section D — Reasoning and Applications

24. (a) Yes. All three sides are equal (AB = PQ, BC = QR, CA = RP), so ΔABC ≅ ΔPQR by **SSS**.
(b) ∠B lies between sides AB and BC; the matching angle lies between PQ and QR, i.e. **∠Q**.

25. (a) No. Checking **three** suitable parts is enough: SSS, SAS, ASA, AAS or RHS.
(b) No, not any three. Three equal angles (AAA) do not guarantee congruence — a small and a large triangle can both have angles 50°, 60°, 70°. Two sides and a non-included angle (SSA) also do not work (see Question 11).

26. (a) AC = PR (the side next to ∠A, so that ∠A is the included angle).
(b) ∠B = ∠Q (the angle at the other end of AB).
(c) ∠C = ∠R (the angle opposite the side AB).

27. (a) AB = AD and BC = DC (sides of a square), AC = AC (common). ΔABC ≅ ΔADC by **SSS** (SAS also works, using ∠B = ∠D = 90°).
(b) Isosceles (AB = BC) and right-angled (∠B = 90°).
(c) ∠BAC = ∠BCA, and together they make 180° − 90° = 90°. So each is **45°**.

28. (a) AB = BA (common), ∠ABC = ∠BAD = 90°, BC = AD (opposite sides of a rectangle). ΔABC ≅ ΔBAD by **SAS**. (RHS can also be used.)
(b) BD = AC = **13 cm** (corresponding parts).
(c) The diagonals of a rectangle are equal in length.

29. (a) They are vertically opposite angles (lines AC and BD cross at O).
(b) OA = OC, ∠AOB = ∠COD, OB = OD. So ΔAOB ≅ ΔCOD by **SAS**.
(c) AB = CD = **25 m**.

30. (a) The three sides of ΔABC are equal. Taking away equal lengths AD, BE, CF from equal sides AB, BC, CA leaves equal lengths: DB = EC = FA.
(b) AD = BE (given), ∠A = ∠B = 60°, AF = BD (from part a). So ΔADF ≅ ΔBED by **SAS**. Hence DF = ED.
(c) In the same way, BE = CF, ∠B = ∠C = 60°, BD = CE, so ΔBED ≅ ΔCFE (SAS) and ED = FE. So DE = EF = FD, i.e. ΔDEF is equilateral.